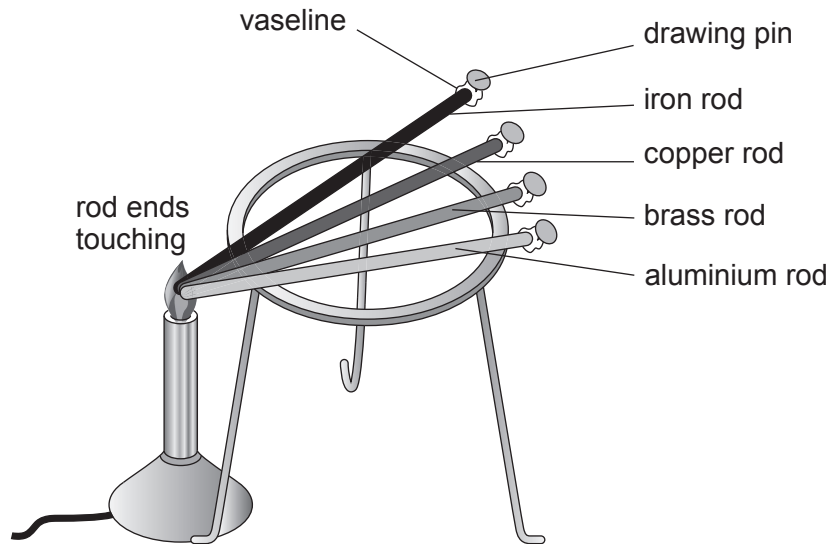


3. Metals are good conductors of heat and electricity.

(a) Three students investigated the thermal conductivity of four metal rods as shown below.



State **two** ways in which the students will make the experimental method valid.

[2]

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(b) The table shows results from the students' investigation together with theoretical values of thermal conductivity.

Metal	Time taken for pin to drop (s)	Thermal conductivity (W/m K)
iron	17	80.4
copper	7	401
brass	12	109
aluminium	5	210

Use the data to explain whether the students' results are as expected.

[3]

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- (c) Explain, in terms of structure and bonding, why the electrical conductivity of sodium chloride is dependent upon its state. [3]

Examiner
only

